JARVIS EP 3 - BY PULSE OF AI

Official Step-by-Step Viewer Guide | Build Your Own Voice AI Assistant Brain

Follow this complete beginner walkthrough from Pulse of AI to set up an isolated virtual environment, connect Google's free neural brain, and run your voice-enabled AI assistant in Visual Studio Code.

Step 1: Project Setup & Virtual Environment

1. **Create Folder:** Right-click desktop → New → Folder → Name it **JARVIS**.
2. **Open in VS Code:** Open Visual Studio Code → File → Open Folder... → Select **JARVIS**.
3. **Create Virtual Environment:** Open integrated terminal (Ctrl + `) and run: python -m venv venv
4. **Activate Environment:** Run .\venv\Scripts\activate (Windows) or source venv/bin/activate (Mac/Linux).

Step 2: Install Required Libraries

Run the following command inside your active terminal:

```
pip install google-genai python-dotenv
```

- **google-genai:** Official Google SDK to connect with high-speed Gemini neural models.
- **python-dotenv:** Safely loads your API key from a hidden .env file.

Step 3: Get Your Free Gemini API Key

1. Visit Google AI Studio and sign in with your Google Account.
2. Click **'Get API key'** in the left sidebar → click **'+ Create API Key'**.
3. In VS Code, create a file named **.env** and paste: GEMINI_API_KEY=your_actual_key_here

Step 4: The brain.py Source Code

Create a new file named **brain.py** and paste the official Pulse of AI production code:

```
import os, sys
from dotenv import load_dotenv
from google import genai
from google.genai import types

load_dotenv()
api_key = os.getenv("GEMINI_API_KEY")
if not api_key:
    sys.exit("Error: GEMINI_API_KEY not found in .env file.")

client = genai.Client(api_key=api_key)

SYSTEM_INSTRUCTIONS = """
You are JARVIS, an elite AI assistant built by Swayam Dhawale, Founder of Pulse of AI.
- Persona: Sharp, concise, witty, loyal, and direct (Tony Stark style).
- Language: Natural Hinglish (Hindi + English mix in Roman script).
- Rules: Never give long essays. Keep replies under 2 crisp sentences and address the user as Sir.
"""

chat = client.chats.create(
    model="gemini-3.6-flash",
    config=types.GenerateContentConfig(system_instruction=SYSTEM_INSTRUCTIONS, temperature=0.7)
)

def talk_to_jarvis(user_message: str):
    response = chat.send_message_stream(user_message)
    for chunk in response:
        print(chunk.text, end="", flush=True)
    print()

if __name__ == "__main__":
    print("=" * 50 + "\n  JARVIS NEURAL CORE ONLINE | BY PULSE OF AI\n" + "=" * 50)
    while True:
        try:
            query = input("\nYou: ").strip()
            if query.lower() in ["exit", "quit", "shutdown"]:
                print("\nJARVIS: Powering down systems. Goodbye, Sir.")
                break
            if query:
```

```
            print("\nJARVIS: ", end="")
            talk_to_jarvis(query)
        except KeyboardInterrupt:
            break
```

Step 5: Run JARVIS & Next Episode Preview

1. Save all files (**Ctrl + S**) → Run in terminal: python brain.py
2. Ask JARVIS: *'Who created you?'* to test the neural persona.
3. **Coming in Episode 4:** We will integrate real-time microphone listening and free Edge-TTS neural voice speech output!
4. Subscribe to Pulse of AI on YouTube for the full open-source JARVIS project series.